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PRACA DYPLOMOWA

Digital Twin-Based Energy Performance Estimation of Refrigeration Appliances Using Minimal Sensor Data

Szacowanie wydajności energetycznej urządzeń chłodniczych w oparciu o cyfrowego bliźniaka z wykorzystaniem minimalnej ilości danych z czujników

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Kraków, 2026

I would like to thank AMC Vibro and ES System K for providing the measurement data used in this thesis. Access to real operational data from refrigeration systems was essential for developing, testing, and comparing the proposed methods.

Streszczenie

Niniejsza praca dotyczy estymacji zużycia energii urządzeń chłodniczych z wykorzystaniem ograniczonego zestawu danych pomiarowych. Głównym celem jest odtworzenie pracy sprężarki, ponieważ czas jej załączenia stanowi podstawę uproszczonej estymacji zużycia energii. W pracy opracowano i porównano dwa podejścia: metodę peak-valley opartą głównie na temperaturze wewnątrz komory chłodniczej oraz model cyfrowego bliźniaka typu gray-box, wykorzystujący temperaturę wewnętrzną, temperaturę otoczenia i dane o stanie sprężarki w procesie uczenia. Metoda peak-valley charakteryzuje się niskimi wymaganiami sprzętowymi i obliczeniowymi oraz może osiągać wysoką dokładność, jednak w niektórych przypadkach wymaga krótkiej kalibracji. Model gray-box jest bardziej złożony, ale zapewnia interpretowalne parametry fizyczne i umożliwia symulację pracy urządzenia w zmienionych warunkach. Metody zweryfikowano na danych z chłodziarek i zamrażarek, wskazując kompromis między prostotą wdrożenia a możliwościami diagnostycznymi modelu.

Abstract

This thesis investigates energy-consumption estimation for refrigeration appliances using minimal sensor data. The work focuses on reconstructing compressor operation, since compressor on-time is the main basis for simplified energy estimation. Two approaches are developed and compared: a peak-valley method based mainly on internal chamber temperature, and a gray-box digital twin model using internal temperature, ambient temperature, and compressor-state data for training. The peak-valley method offers low hardware and computational requirements and can achieve high accuracy, but may require a short calibration period when temperature extrema are shifted relative to compressor transitions. The gray-box method is more complex and computationally demanding, but provides physically interpretable parameters and enables simulation under changed operating conditions. The methods are evaluated on refrigerator and freezer datasets, showing the trade-off between low-cost deployment and model-based diagnostic capability.

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Lista symboli

Model parameters

- R Thermal resistance of the cabinet enclosure (K/W)
- C Effective thermal capacitance of the refrigerated chamber (J/K)
- G Compressor heat-removal coefficient (W/K)
- P_{on} Power draw when the compressor is operating (W)
- P_{off} Power draw when the compressor is off (W)

States and signals

- $T(t)$ Internal cabinet temperature ($^{\circ}\text{C}$)
- $T_a(t)$ Ambient temperature ($^{\circ}\text{C}$)
- $u(t)$ Compressor state indicator, where 1 denotes on and 0 denotes off
- $q_c(t)$ Analogous capacitor charge in the gray-box thermal model
- $V_c(t)$ Analogous capacitor voltage representing internal cabinet temperature
- $I(t)$ Analogous current representing heat removal by the evaporator
- $x(t)$ State variable of the gray-box state-space model
- $y(t)$ Simulated internal temperature output of the gray-box model

Derived quantities

- A State matrix of the gray-box state-space model

B Input matrix of the gray-box state-space model

C_s Output matrix of the gray-box state-space model

D Feedthrough matrix of the gray-box state-space model

$P = [R, C, G]$ Parameter vector estimated in the gray-box model

Δk Timing offset between a detected extremum and measured compressor transition

t_{on} Total compressor on-time in a selected segment (s)

$t_{\text{on,pred}}$ Predicted compressor on-time in a selected segment (s)

$t_{\text{on,meas}}$ Measured compressor on-time in a selected segment (s)

Optimization and validation

APE Absolute percentage error of compressor on-time estimation

MAPE Mean absolute percentage error of compressor on-time estimation

RMSE Root-mean-squared error between simulated and measured temperature

GA Genetic algorithm used for gray-box parameter identification

1. Introduction

Reliable estimation of refrigeration-appliance energy performance is difficult when direct electrical measurements are unavailable or impractical. The work focuses on recovering compressor operation from limited thermal measurements, because compressor on-time is the dominant factor in simplified energy-consumption estimation. Two complementary approaches are considered: a minimal-sensor peak-valley method and a gray-box digital twin method.

1.1. Background and Motivation

Refrigeration appliances are widely used in domestic, commercial, and industrial applications. Their energy consumption is strongly related to the operation of the compressor and to the thermal interaction between the insulated chamber and the surrounding environment. Reliable estimation of their energy performance is therefore important for appliance validation, monitoring, and diagnostics.

Traditional energy assessment usually requires direct electrical measurements or a relatively rich sensor setup. This thesis investigates whether useful estimation can be obtained from minimal sensor data, ideally only the temperature measured inside the refrigerated chamber. This constraint is important for low-cost monitoring and for practical deployment on existing appliances.

1.2. Research Problem

The main problem addressed in this thesis is the estimation of compressor operation and energy consumption of refrigeration appliances from indirect thermal measurements. If the compressor state can be inferred from the temperature signal, the total compressor on-time and the corresponding energy consumption can be estimated using a simplified two-state power model. This is useful in situations where electrical power measurements are not available for the full operating period, but temperature data is available from the appliance controller or from an added low-cost sensor.

1.3. Research Objective

The objective of this thesis is to develop and validate methods for estimating compressor state and energy performance of refrigeration appliances using minimal measured data. Two approaches are considered: a signal-processing peak-valley method and a gray-box digital twin method. The first approach prioritizes simplicity and low hardware requirements, while the second prioritizes physical interpretation and predictive capability.

1.4. Research Questions

The thesis is guided by the following research questions:

- Can compressor duty cycle be inferred reliably from internal temperature measurements?
- Under what conditions is a simple peak-valley method sufficient?
- Does a gray-box digital twin improve prediction capability or generalization?
- How much labeled calibration or training data is required?
- What trade-off exists between sensor requirements, calibration effort, and accuracy?

1.5. Contributions

The main contributions of this thesis are:

- Development of a peak-valley method for inferring compressor state from internal temperature data.
- Formulation of a calibration protocol for correcting systematic timing offsets between extrema and compressor switching events.
- Application of a gray-box digital twin model for compressor-state and energy-performance estimation.
- Comparative evaluation of both approaches on measured refrigerator and freezer datasets.
- Analysis of the trade-off between minimal sensing, calibration effort, and estimation accuracy.

2. State of the Art

Refrigeration energy performance, fault detection, and digital-twin modelling are closely connected because the same measured signals can be used for monitoring, diagnostics, and parameter identification. Existing work shows strong results for machine-learning and virtual-sensing methods, but often at the cost of additional sensors, large datasets, or limited physical interpretability. This motivates a comparison between a simple temperature-based estimator and a physically interpretable gray-box model.

2.1. Refrigeration Energy Performance and Thermal Behaviour

Refrigeration systems are essential for preserving perishable goods such as food and pharmaceuticals. Their energy performance depends on the interaction between the refrigerated chamber, the ambient environment, the thermal load, and the vapour-compression refrigeration cycle. The physical basis of heat transfer and air flow in domestic refrigerators is described by Laguerre [1], who shows that internal temperature dynamics are governed by coupled heat-transfer mechanisms, air circulation, wall insulation, and the distribution of stored products. This supports the use of reduced-order thermal models when the objective is not to reproduce every local phenomenon, but to capture the dominant temperature response of the appliance.

The influence of thermal load is especially important when estimating compressor activity from temperature data. Belman-Flores et al. [2] experimentally studied the thermal and energy behaviour of a domestic refrigerator under different load conditions and showed that load changes significantly affect energy consumption and cycling behaviour. This is directly relevant to this thesis because thermal load changes may alter the time delay between compressor switching and measured chamber-temperature extrema. Hessami [3] also demonstrated that refrigerator energy rating can be estimated using heat-transfer measurements and computer simulations, which supports the broader idea that energy performance can be inferred through thermal modelling rather than only through direct long-term electrical measurements.

Early refrigeration modelling also includes data-driven thermodynamic prediction. Kizilkan [4] used artificial neural networks to model a variable-speed compressor refrigeration system based on limited experimental data, showing that data-driven methods can predict system behaviour under different compressor frequencies and cooling loads. However, such methods generally do not provide direct physical parameter interpretation, which limits their usefulness when the objective includes system identification or fault interpretation.

2.2. Predictive Maintenance and Anomaly Detection

Predictive maintenance aims to detect abnormal operation before failure occurs, reducing downtime, energy waste, and maintenance cost. A common approach is to process raw sensor signals, extract statistical features, and apply anomaly-detection or classification methods. Rabatel et al. [5] demonstrated this workflow for preventive maintenance using monitoring sensor data. Their work highlights the importance of data cleaning, feature extraction, and statistical filtering, but also illustrates a limitation of purely statistical methods: detected anomalies are not always easy to connect to physical causes.

The evaluation of anomaly detection in predictive maintenance is also non-trivial. Carrasco et al. [6] proposed an evaluation framework for temporal unsupervised anomaly-detection algorithms, including interval-based generalisation of positive and negative instances and a metric that rewards early detection. This is important because maintenance-oriented algorithms should not only classify isolated samples correctly but should also detect abnormal intervals early enough to be useful. For refrigeration systems, this suggests that evaluation should consider operational cycles, time intervals, and practical decision thresholds rather than only pointwise classification accuracy.

Recent surveys confirm that anomaly detection and predictive maintenance are dominated by data-driven techniques. Kamat and Sugandhi [7] reviewed anomaly-detection methods for Industry 4.0, covering techniques from statistical process control to deep-learning autoencoders. Forbicini et al. [8] surveyed time-series analysis in compressor-based machines and distinguished tasks such as fault detection, fault prediction, forecasting, and change-point detection. These surveys emphasise the increasing use of LSTM networks, convolutional models, support-vector machines, and tree-based methods. At the same time, they point to recurring problems: limited public benchmark datasets, high dependence on data quality, and the need for methods that remain interpretable and deployable in real industrial conditions.

2.3. Machine Learning and Virtual Sensing in Refrigeration

Several recent studies apply machine learning to refrigeration monitoring and fault detection. Quispe-Astorga et al. [9] developed a sensor-based machine-learning classification system for cooling-unit fault detection and diagnosis. Their setup used pressure, current, voltage, and temperature sensors with embedded data acquisition and edge processing. The reported classification performance is high, but the approach requires a relatively complex hardware setup.

Potharaju et al. [10] proposed a two-step predictive-maintenance method in which Isolation Forest is first used to generate pseudo-labels from unlabeled environmental sensor data, followed by supervised classifiers such as Random Forest, multilayer perceptron, and AdaBoost. This type of approach can achieve high accuracy, but it depends on large datasets, model training, and computational resources that may not be available in a low-cost refrigeration monitoring scenario.

More refrigeration-specific work attempts to reduce the sensor burden while retaining predictive capability. Farahani et al. [11] used virtual sensors and tree-based machine-learning models for fault detection and diagnosis in supermarket CO₂ refrigeration systems, combining physical relationships with machine-learning classification and SHAP-based interpretation. Wijethunga et al. [12] used categorical gradient boosting with advanced thresholding for leak detection in supermarket refrigeration, relying on features such as coefficient of performance, subcooling, superheat, mass flow, and compression ratio. Alcaraz et al. [13] investigated machine-learning-based virtual sensors for reduced energy consumption in frost-free refrigerators.

These studies show that machine-learning and virtual-sensing methods are effective tools for refrigeration monitoring. However, they often require several measured variables, large datasets, or complex feature engineering. This motivates the question addressed in this thesis: whether useful compressor-state and energy-performance estimation can be achieved with a much smaller measurement set, ideally based mainly on the internal chamber temperature.

2.4. Gray-Box Thermal Modelling and Digital Twins

Gray-box models provide a compromise between first-principles modelling and black-box machine learning. Their structure is based on physical reasoning, while unknown parameters are identified from measured data. Madsen and Holst [14] introduced continuous-time thermal modelling using thermodynamic analogies to electrical RC networks. This approach

is useful for refrigeration because thermal resistance, thermal capacitance, and heat input or extraction terms can be interpreted physically.

The parameter-estimation foundation for stochastic gray-box models was presented by Kristensen et al. [15]. Their maximum-likelihood and Kalman-filter-based framework explains how model parameters can be estimated from noisy time-series data. Although this thesis uses a more implementation-oriented optimisation procedure, the same general principle applies: measured temperature trajectories are used to identify physically meaningful parameters of a reduced-order thermal model.

Gray-box modelling has been applied directly to refrigeration systems. Sossan et al. [16] modelled a household refrigeration unit using time-series data for demand-side management. Their work is particularly relevant because it uses a thermodynamic state-space structure close to the type of model considered in this thesis and discusses parameter-identifiability challenges. Leerbeck et al. [17] developed gray-box models for supermarket refrigeration cabinets, showing that physically interpretable models can be used for system identification and monitoring. Huang et al. [18] proposed a gray-box modelling approach for ultra-low-temperature freezing chambers, demonstrating the use of more complex thermal models for digital operations.

The immediate methodological background for this thesis is the digital-twin approach developed by Pedzik, Suchon, and Barszcz [19]. That work bridges gray-box modelling and machine learning for refrigeration-system identification and predictive maintenance. It demonstrates that estimated physical parameters can be connected to changes in system behaviour and that a digital twin can support anomaly detection and model-based interpretation. The present thesis extends this direction toward energy-performance estimation and compressor-state prediction using minimal sensor data.

2.5. Compressor State Estimation from Minimal Sensor Data

The literature contains many studies on refrigeration modelling, fault detection, and virtual sensing, but comparatively fewer works focus specifically on inferring compressor state or compressor on-time from minimal measurements. This is an important distinction. In the present thesis, the final energy estimate is based on compressor operation: if the compressor state is known or estimated, the appliance power can be approximated by a two-state model with P_{on} during compressor operation and P_{off} during idle operation.

Temperature-based compressor-state estimation is attractive because internal temperature is easy to measure and is often already available. The cyclic operation of a hysteresis-controlled

appliance produces peaks and valleys in the chamber-temperature signal. These extrema can be used as indirect indicators of compressor switching events. However, the relation is not exact. Sensor placement, thermal inertia, ambient variation, door openings, and changes in thermal load can create timing offsets between the true compressor transitions and the observed temperature extrema. Therefore, a practical minimal-sensor method must include not only signal processing but also validation and calibration.

Gray-box digital twins offer a second route to compressor-state estimation. Instead of detecting extrema directly, a calibrated thermal model can simulate the internal temperature and compressor-control logic under a given ambient-temperature profile. This requires more calibration data than a simple peak-valley method, but it provides additional capabilities: physical parameter interpretation, simulation of future operation, and potential use in predictive maintenance.

2.6. Research Gap

The reviewed literature shows three main trends. First, refrigeration energy performance is strongly governed by thermal load, insulation, and compressor cycling, which supports the use of thermal models for energy estimation. Second, predictive-maintenance research increasingly uses machine learning and virtual sensing, often achieving high accuracy but requiring many sensors, large datasets, and limited physical interpretability. Third, gray-box models provide interpretable physical parameters and are well suited to digital twins, but they may suffer from parameter-identifiability issues and often require measurements that are not always available in low-cost deployments.

The specific gap addressed by this thesis is the practical estimation of compressor operation and energy performance of refrigeration appliances using minimal sensor data. The thesis compares two complementary approaches: a peak-valley method based mainly on internal temperature and a gray-box digital twin method based on physical thermal modelling and a short labelled training period. The contribution is not only the implementation of these methods but also their evaluation in terms of accuracy, calibration effort, sensor requirements, interpretability, and practical deployability.

3. Research Object and Problem Formulation

The considered system is a hysteresis-controlled refrigeration appliance whose compressor operation is reflected indirectly in the internal temperature trajectory. The engineering problem is formulated as the reconstruction of a binary compressor mask from available measurements, followed by the estimation of compressor on-time and energy consumption using a simplified two-state power model.

3.1. Refrigeration System Description

The research object is a refrigeration appliance, such as a refrigerator or freezer, consisting of an insulated chamber, compressor, condenser, evaporator, expansion element, and controller. The appliance exchanges heat with the ambient environment and removes heat from the chamber through the refrigeration cycle.

From the point of view of this thesis, the most important part of the appliance is the refrigerated chamber. It is a thermally insulated volume in which products are stored or displayed, and whose internal air temperature is controlled within a specified range. Heat enters the chamber through the cabinet walls, door seals, air exchange during door openings, and thermal loads introduced with products. The refrigeration system compensates for these heat gains by transporting thermal energy from the chamber to the surrounding environment.

The cooling system is based on a vapor-compression cycle. In the evaporator, the refrigerant absorbs heat from the air and surfaces inside the chamber. The compressor then increases the pressure and temperature of the refrigerant vapor, allowing heat to be rejected to the ambient environment in the condenser. After condensation, the expansion element lowers the refrigerant pressure and temperature before it returns to the evaporator. In this way, mechanical work supplied to the compressor enables heat transfer from a lower-temperature region inside the cabinet to a higher-temperature ambient environment [19].

For the methods developed in this thesis, the full thermodynamic complexity of the refrigeration cycle is not modeled directly. Instead, the appliance is treated as a controlled thermal

system whose observable behavior is dominated by the internal temperature trajectory and compressor switching. This abstraction is sufficient for estimating compressor operation and energy use from minimal sensor data, while still preserving the physical interpretation needed for the gray-box model.

3.2. Control Logic and Operating Principle

The considered appliances are assumed to use hysteresis control. The compressor switches on when the internal temperature exceeds an upper threshold and switches off when the temperature falls below a lower threshold. This produces cyclic behavior in both the internal temperature signal and the compressor state. The control system is described in diagram 3.1, and an example of simulation results is shown in Figure 3.2.

3.3. Available Measurements and Datasets

The available measurements may include internal temperature, ambient temperature, compressor state, and electrical power. The methods in this thesis are designed around minimal sensing, with the peak-valley method primarily using internal temperature and the gray-box method using a short labeled period with internal temperature, ambient temperature, and compressor state.

The datasets used in the evaluation are shown in Table 3.1. Example of data in Figure 3.3.

Table 3.1.: Datasets used in the thesis.

Dataset	Appliance	Total duration	Usable duration	Available signals
1	Refrigerator	3.2 h	3.2 h	Internal temperature, compressor state
2	Freezer	1 month	546–567 h	Internal temperature, ambient temperature, compressor state
3	Refrigerator	75 h	40 h	Internal temperature, ambient temperature, compressor state

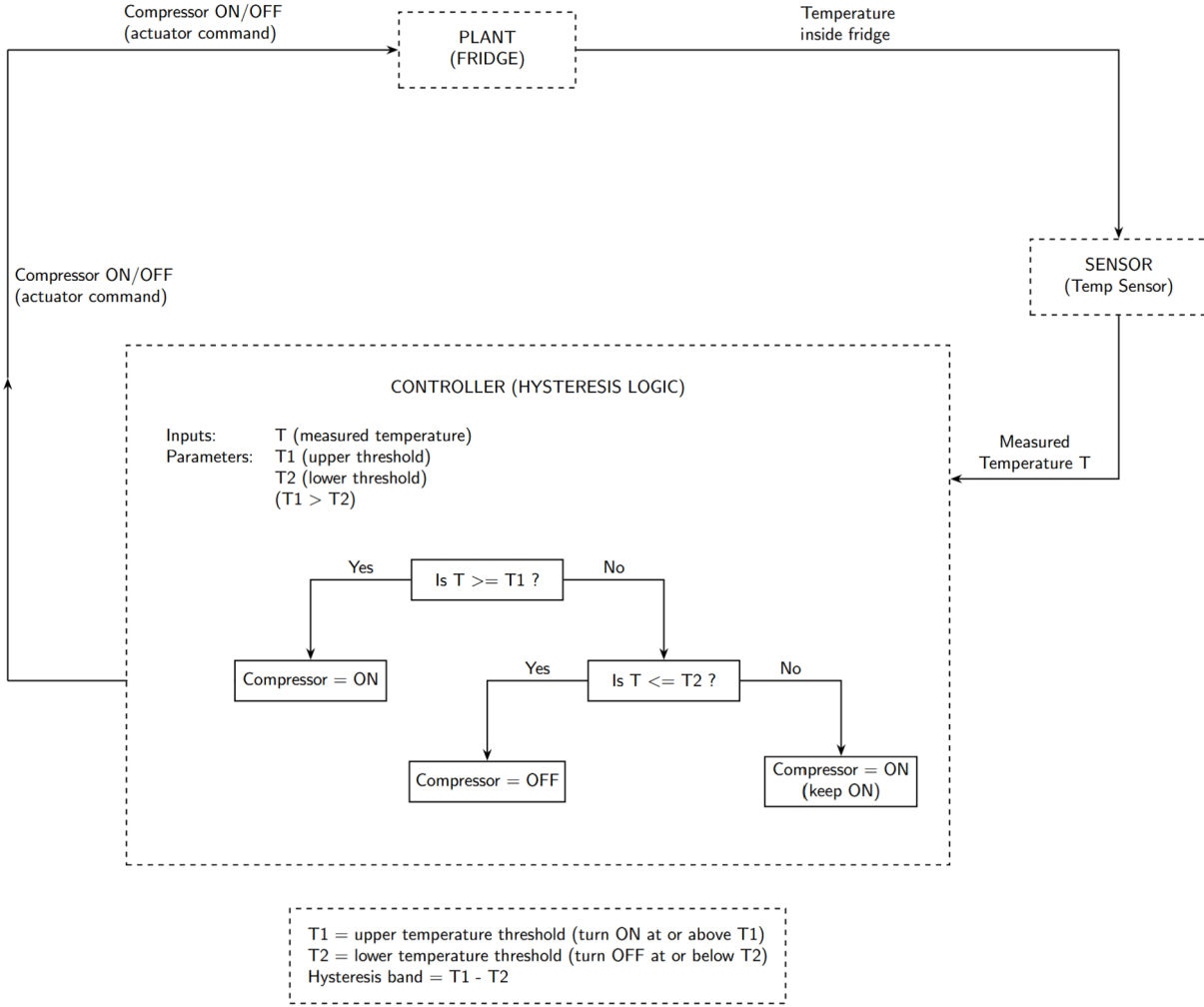


Figure 3.1.: Diagram describing the fridge control logic

3.4. Problem Statement

The goal is to estimate a binary compressor mask from measured temperature data. The compressor mask indicates when the compressor is operating and when it is off. Once this mask is known, the total compressor on-time can be calculated for any selected segment of data. If the compressor-on and compressor-off power levels are known or assumed, the same mask can be used to estimate energy consumption.

The problem is challenging because the internal temperature does not change instantaneously when the compressor switches. The measured temperature response is delayed by thermal

inertia, sensor placement, air mixing, and the thermal mass of stored products. Therefore, the thesis evaluates both a direct signal-based method and a model-based method.

3.5. Power Model Assumptions

The appliance power is modeled as a two-state function. This approximation assumes that the dominant change in electrical power is caused by compressor operation. Auxiliary loads such as electronics, lighting, or fans are treated as part of the off-state power unless stated otherwise.

$$P(t) = \begin{cases} P_{\text{on}}, & \text{compressor on,} \\ P_{\text{off}}, & \text{compressor off.} \end{cases} \quad (3.1)$$

The idle power P_{off} may be negligible or may include electronics, lighting, fans, and other auxiliary components. The transition between the two power states is assumed to be instantaneous with respect to the compressor state.

3.6. Evaluation Metrics and Validation Procedure

The main evaluation quantity is the error in the total compressor on-time. This metric is used because the total on-time directly determines the compressor-energy component in the simplified two-state power model. The absolute percentage error for a segment is defined as:

$$\text{APE} = 100\% \cdot \frac{|t_{\text{on,pred}} - t_{\text{on,meas}}|}{t_{\text{on,meas}}}. \quad (3.2)$$

For multiple segments, the mean absolute percentage error is:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \text{APE}_i. \quad (3.3)$$

In this thesis, a segment with MAPE above 10% is treated as an unacceptable result unless stated otherwise.

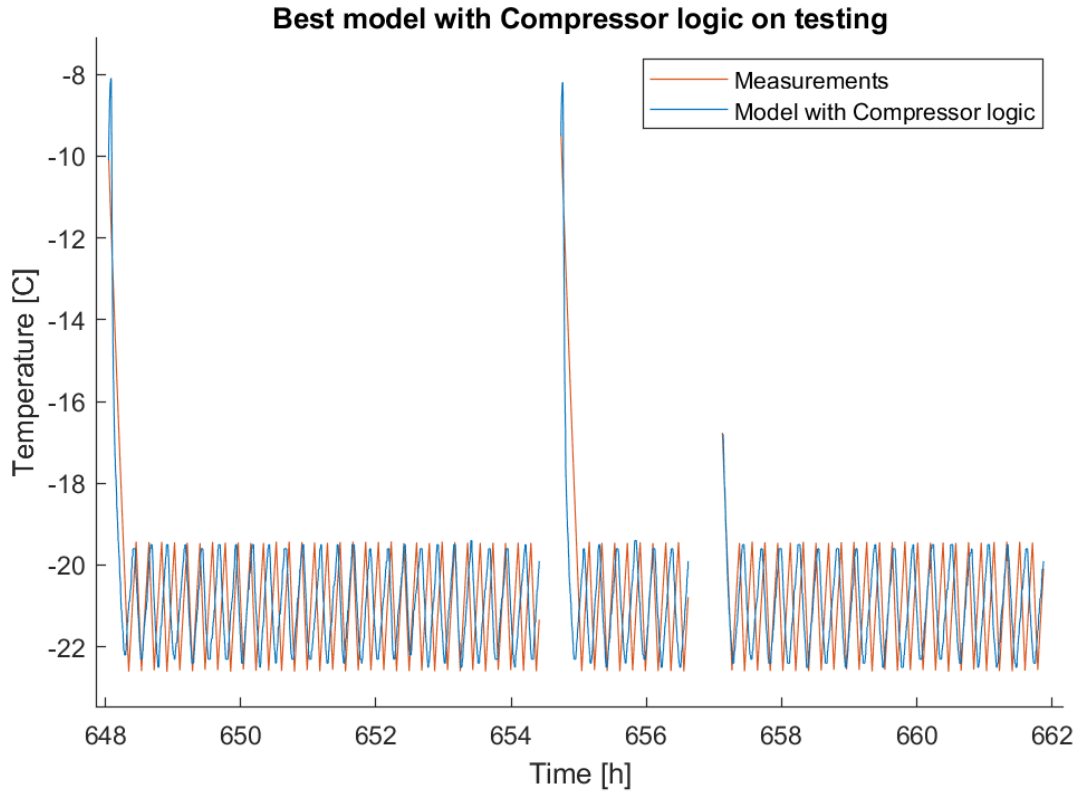


Figure 3.2.: Dynamic temperature simulation of a discrete-time state-space model with hysteresis control using the parameter set identified from the training dataset compared against independent test data.

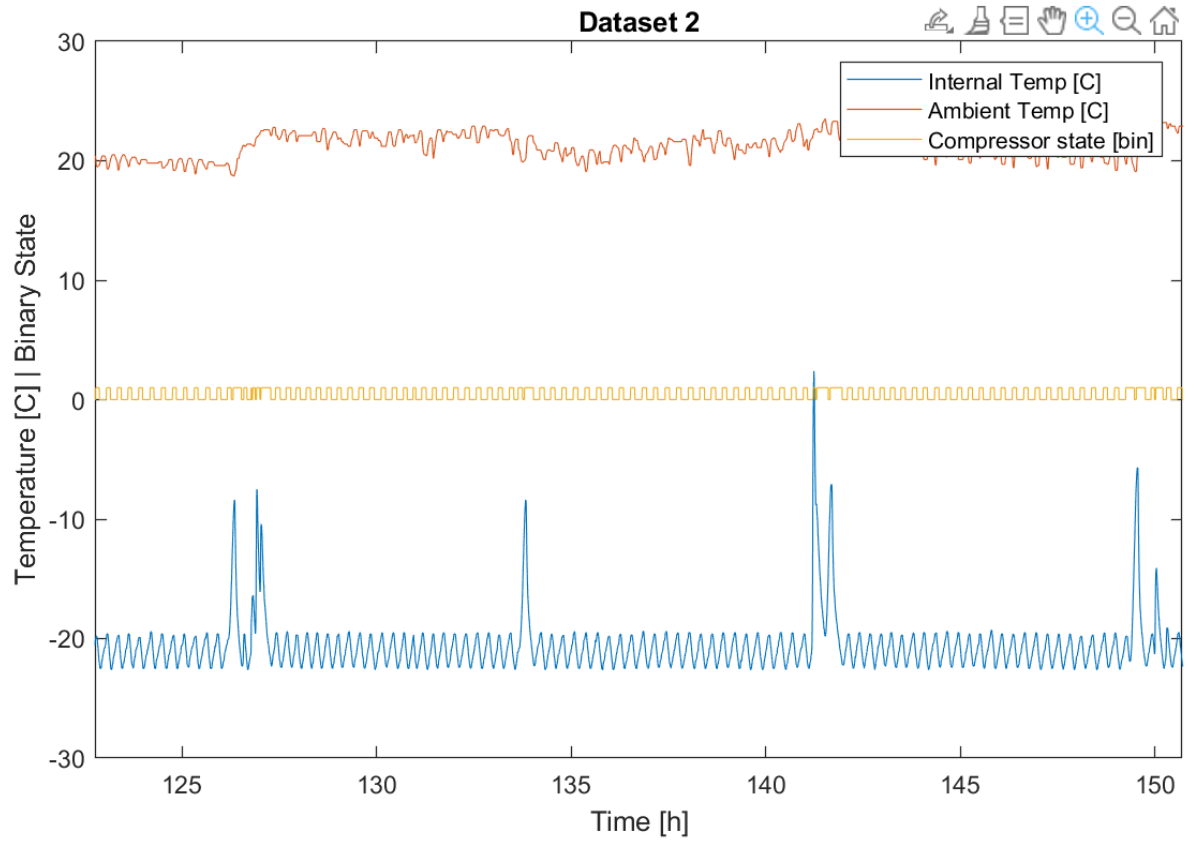


Figure 3.3.: Plot of Dataset3 over time for a short period of time.

4. Data Preparation and Preprocessing

Reliable compressor-state estimation requires temperature signals that are continuous, physically plausible, and not dominated by measurement faults or non-stationary disturbances. The detailed data-processing protocol used as the basis for this work was described in Chapter 6 of [20]. That work presents the full procedure for time-series data analysis, measurement-error correction, anomaly detection, and preparation of stationary data chunks for gray-box parameter estimation. The preprocessing used here follows the same general logic before applying the peak-valley and gray-box methods.

4.1. Data Quality Assessment

Before applying either estimation method, the measured data should be checked for missing values, outliers, unrealistic temperature changes, flat periods, and corrupted segments. Plotting the signals and inspecting temperature histograms are recommended first steps because both proposed methods depend on the shape and continuity of the temperature trajectory. This step is especially important for temperature-based compressor estimation because a short measurement error can be incorrectly interpreted as a peak, valley, or abnormal compressor cycle.

4.2. Handling Missing and Faulty Measurements

Short and sparse interruptions may be linearly interpolated if the degradation is limited. Figure 4.1 shows an example of the internal refrigerator temperature, where a short faulty interval is reconstructed to preserve a continuous signal for later analysis. Longer corrupted periods should be removed from the analysis instead of being interpolated, because they may distort compressor-cycle timing. The same preprocessing rules should be applied consistently when comparing the peak-valley and gray-box methods.

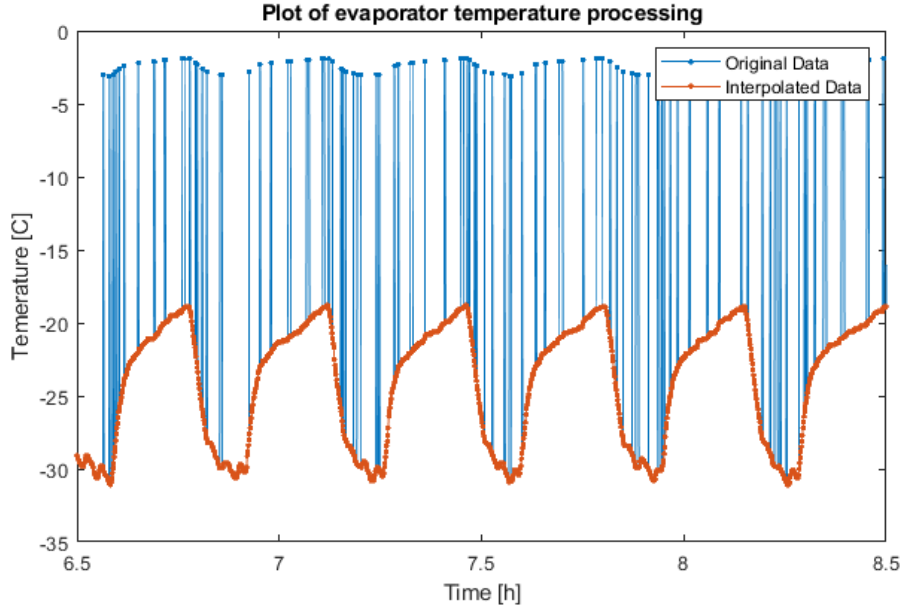


Figure 4.1.: Linear interpolation of a short faulty interval in the internal refrigerator temperature signal.

4.3. Segment Extraction and Anomaly Removal

After anomaly removal, the data is divided into continuous segments, referred to as chunks. A chunk is a time interval in which the measurement is continuous and suitable for further processing. This segmentation is necessary because removed disturbances create gaps in the signal, and directly joining separated parts would introduce artificial phase discontinuities. Since the beginning and end of chunks may be unreliable, very short chunks are excluded from the evaluation.

4.4. Minimum Chunk Length Analysis

A convergence analysis was performed to choose the minimum chunk length. Figure 4.2 shows how the number of chunks with MAPE above 10% changes as the minimum accepted chunk length is increased. This is important because the start and end of each chunk are often distorted and may not represent stable operation. Figure 4.3 shows the corresponding loss of usable data. If the minimum chunk length is set too high, many otherwise useful measurements are discarded. Based on this trade-off, a minimum duration of 2 h was selected because it reduced the number of high-error segments while excluding only a small part of the total data.

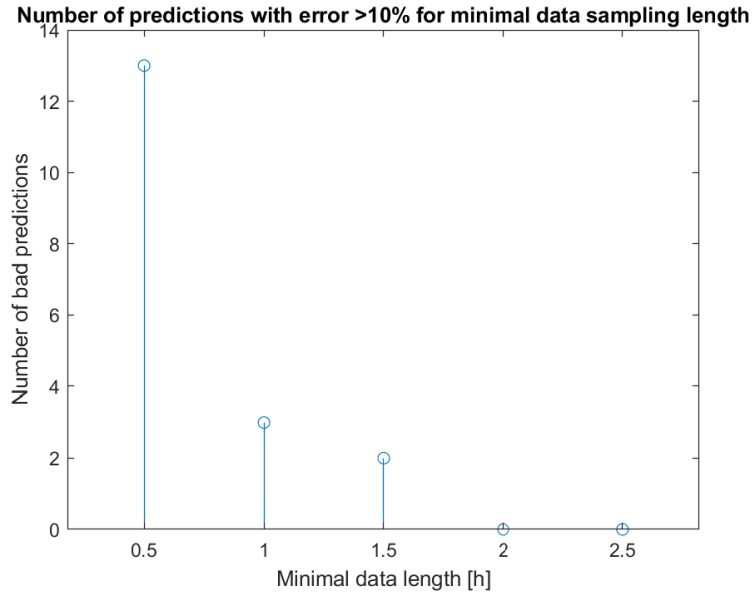


Figure 4.2.: Number of chunks with MAPE above 10% as a function of the minimum accepted chunk length.

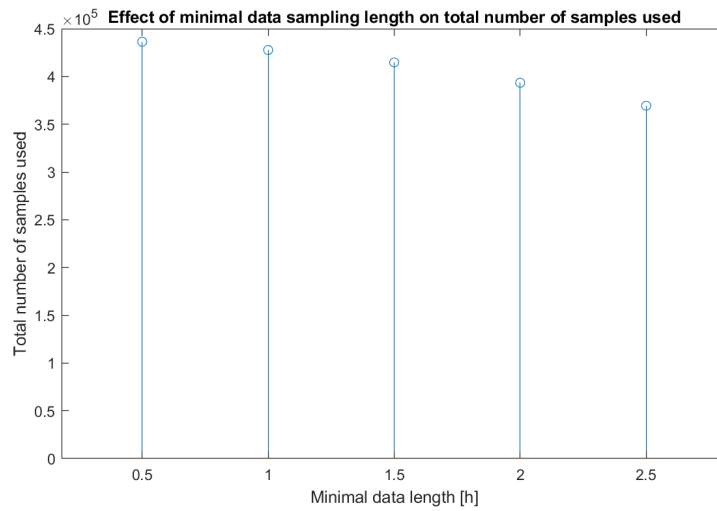


Figure 4.3.: Effect of the minimum accepted chunk length on the total amount of usable data.

5. Peak-Valley Compressor State Estimation

The peak-valley method is a lightweight signal-processing approach for estimating compressor operation from the cyclic shape of the internal temperature signal. Local maxima and minima are treated as indirect markers of compressor transitions, and a calibration step is used when these extrema are systematically shifted relative to the measured compressor state. The method is designed for low-cost deployment where only temperature measurements are available during normal operation.

5.1. Method Overview

The peak-valley method converts a measured internal temperature signal into an estimated binary compressor state. It is based on the observation that compressor switching produces cyclic local extrema in the internal temperature signal. In a hysteresis-controlled appliance, the internal temperature typically rises while the compressor is off and decreases after the compressor turns on. This creates a repeated sequence of local maxima and minima that can be used as indirect markers of compressor transitions.

5.2. Peak and Valley Detection

Local temperature peaks are identified first. A peak is defined as a local maximum that is sufficiently separated in time from other peaks and has sufficient prominence. The temperature signal is then inverted, and the same detection procedure is applied to identify valleys, which correspond to local minima in the original signal. Figure 5.1 shows the resulting detected peaks and valleys on a sampled refrigerator temperature signal.

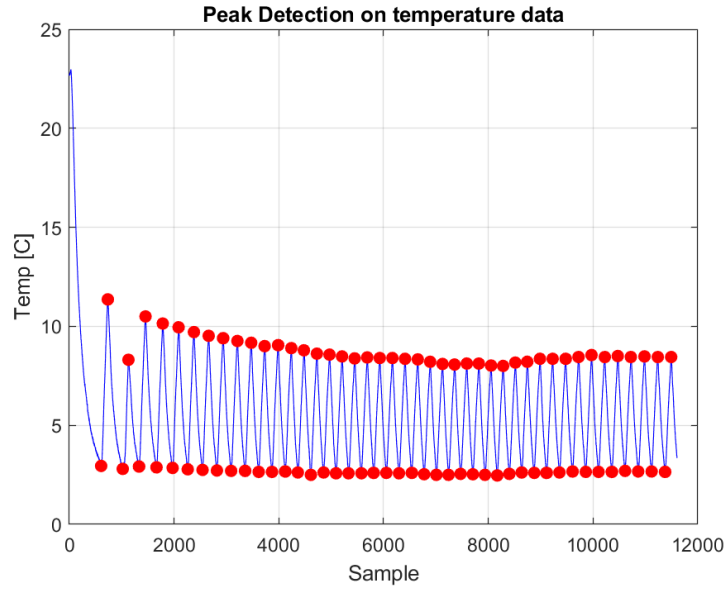


Figure 5.1.: Detected peaks and valleys in a sampled refrigerator temperature signal.

5.3. Compressor Mask Generation

The detected peaks and valleys are ordered in time. Each peak is interpreted as the beginning of a compressor-on period, and each following valley is interpreted as the beginning of a compressor-off period. The interval from a peak to the next valley is labeled as compressor on, while the interval from a valley to the next peak is labeled as compressor off. This binary sequence is referred to as the compressor mask. Figure 5.2 compares the compressor mask estimated from the peak-valley method with the measured compressor state.

This assignment should be verified for each appliance because the physical interpretation may depend on sensor placement and system dynamics.

5.4. Calibration of Timing Offsets

Some datasets show timing shifts between detected extrema and measured compressor transitions. In favourable cases, such as the example in Figure 5.4, the prediction may be shifted in time, but the average delay at compressor start-up and shutdown is similar. The total on-time error can then remain low; in the shown segment, the APE was 1.3%, and on a month-long freezer dataset, the total APE was 2.30% over 546 h of usable data. In less favourable cases, the shift is asymmetric: the delay at compressor start-up differs from the delay at shutdown. Figure 5.3 illustrates the basic compressor-mask assumption used before

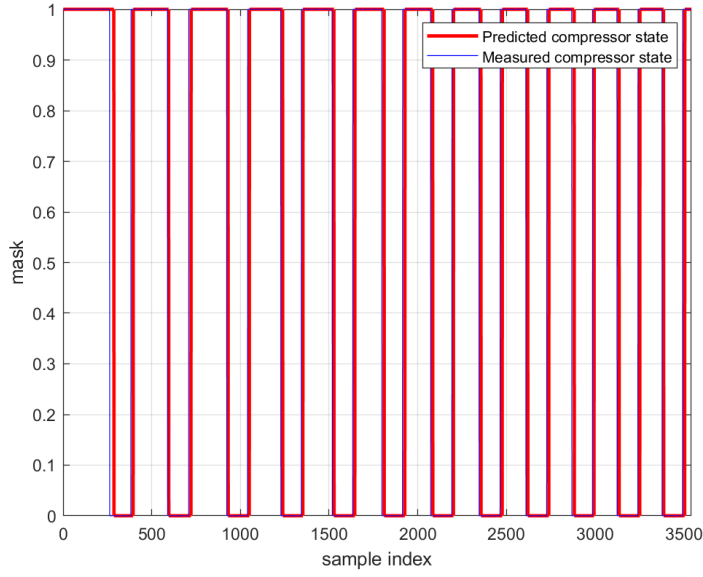


Figure 5.2.: Compressor state estimated by the peak-valley method compared with the measured compressor state.

calibration, where a peak marks compressor turn-on and a valley marks compressor turn-off. When this assumption produces unequal timing errors, the estimated compressor cycles may become systematically too short or too long, which can substantially increase on-time error.

The calibration protocol uses a short labeled period with a known compressor state. Peaks and valleys are matched to the nearest measured compressor transitions. For each accepted match, the offset is computed as:

$$\Delta k = k_{\text{transition}} - k_{\text{extremum}}. \quad (5.1)$$

Offsets are pooled separately for peaks and valleys. A robust central value, such as the median, is used to obtain the final integer peak and valley offsets. These offsets are then applied to unlabeled data.

5.5. Calibration Protocol

The general protocol is:

1. Select a short calibration period with internal temperature and measured compressor state.

2. Run the peak-valley method on the calibration period.
3. Evaluate the on-time error.
4. If the error exceeds the accepted threshold, estimate peak and valley offsets.
5. Apply the calibrated offsets to later measurements without compressor-state labels.

The draft assumes that approximately 6–12 h of labeled data may be sufficient for this calibration step.

Figures 5.5 and 5.6 show the effect of applying this calibration. Before calibration, the predicted compressor cycle is shorter on average than the measured cycle, producing 15.2% APE in the shown segment. After applying the estimated offsets, the segment error decreases to 4.8% APE. On the full 40 h dataset, calibration reduced the peak-valley MAPE from 12.90% to 3.36%.

5.6. Method Parameters

The parameters of the method include peak prominence, minimum peak distance, maximum matching distance, peak offset, valley offset, and minimum accepted chunk length. These parameters control how sensitive the algorithm is to small oscillations, noise, and short corrupted fragments. They should be reported for every evaluated dataset because changing them may alter both the detected number of cycles and the estimated total compressor on-time.

5.7. Advantages and Limitations

The peak-valley method is simple and can operate primarily from internal temperature data. This makes it attractive for low-cost monitoring, embedded implementation, and retrofitting existing appliances with minimal additional hardware. Its main limitation is sensitivity to asymmetric thermal response, sensor placement, noise, and disturbances such as door openings or changing thermal load. For this reason, calibration with a short compressor-state measurement period is recommended when reliable on-time estimation is required.

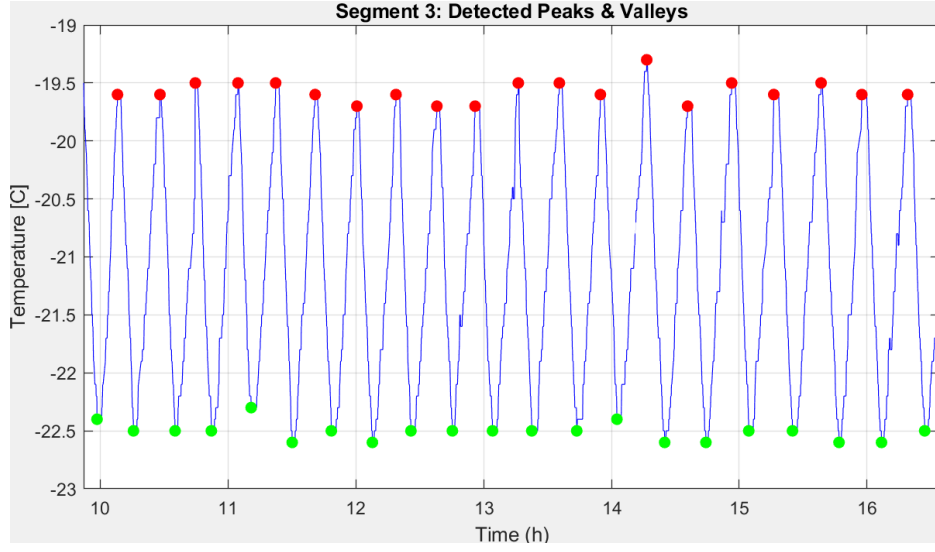


Figure 5.3.: Compressor-mask generation from detected temperature peaks and valleys before timing calibration.

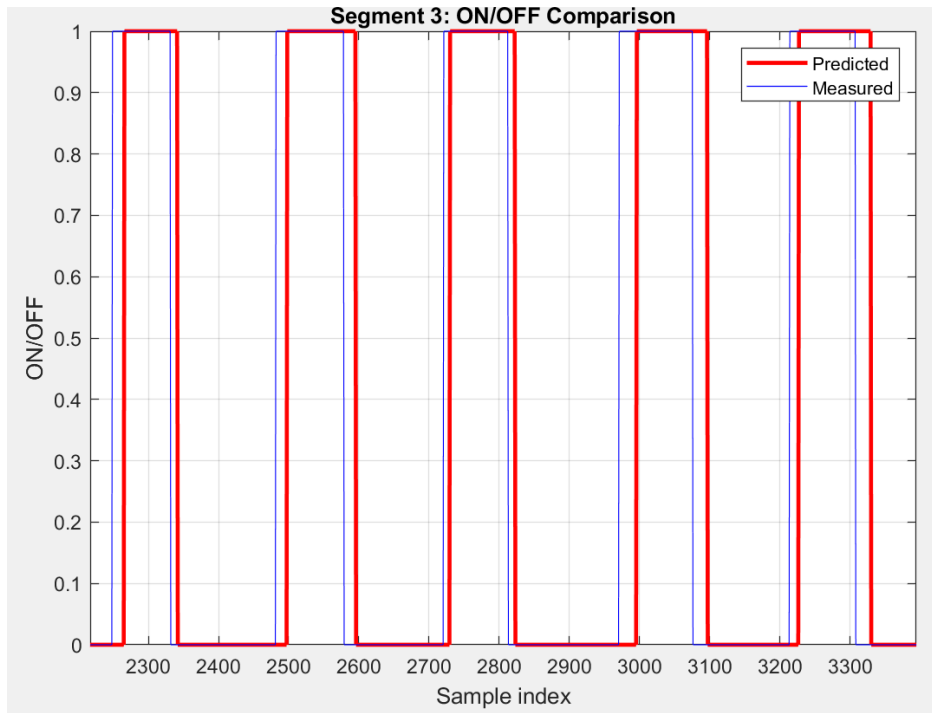


Figure 5.4.: Example of a time-shifted prediction with similar start-up and shutdown delays, resulting in low total on-time error.

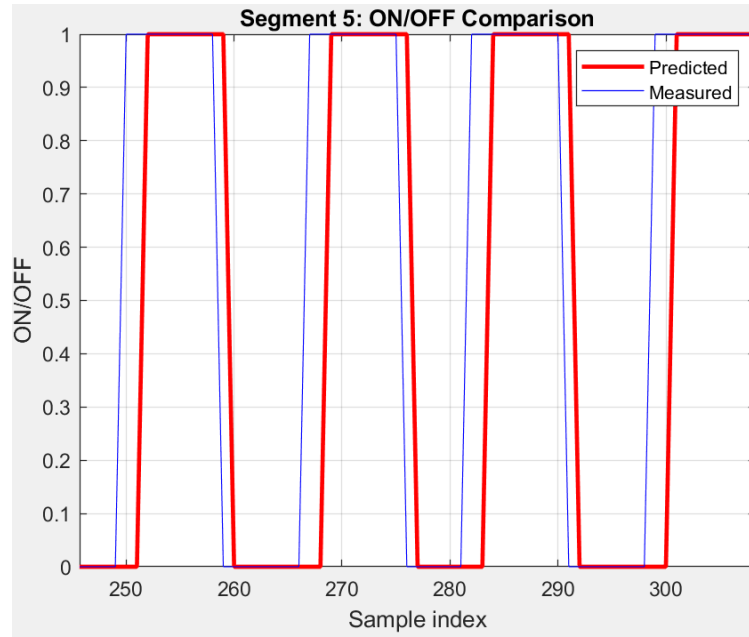


Figure 5.5.: Segment before timing calibration, with predicted compressor cycles shorter than the measured cycles.

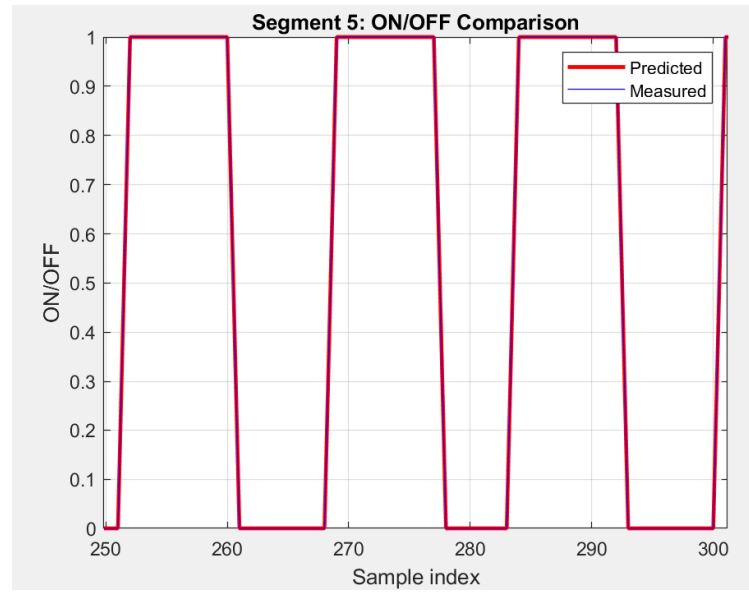


Figure 5.6.: The same segment after timing calibration, showing improved alignment between predicted and measured compressor states.

6. Gray-Box Digital Twin Modeling

The gray-box method represents the refrigeration appliance as a simplified physical thermal system rather than only as a temperature signal. The model combines an RC-type thermal analogy, parameter identification, and hysteresis-based compressor control to simulate internal temperature and compressor operation. Compared with the peak-valley method, it requires more data and computation, but provides physically interpretable parameters and broader diagnostic capability.

6.1. Model Overview

The gray-box method uses a simplified physical model of the refrigeration appliance. The model input is the compressor state and ambient temperature, while the model output is the simulated internal temperature. The model parameters represent physical properties such as thermal resistance, thermal mass, and the compressor heat-removal coefficient. The digital twin used in this thesis follows the same structure as the model described in [19]: a thermodynamic cabinet model represented through an electrical RC analogy and fitted to measured refrigeration-system data.

6.2. Thermal Model Structure

The model, depicted in Figure 6.1, is based on the standard thermal-electrical analogy. Temperature is represented by voltage, heat flow by current, thermal resistance by an electrical resistor, and thermal capacitance by a capacitor. In this interpretation, the refrigerated cabinet, including the internal air, stored goods, and internal surfaces, is represented by an effective heat capacity. Heat exchange with the surroundings is represented by a thermal resistance between the cabinet interior and ambient air. The evaporator acts as a heat sink that removes energy from the cabinet when the compressor is operating.

The compressor, condenser, and expansion valve are not modeled as separate dynamic components. Their influence on the cabinet is represented through the heat-removal input applied at the evaporator. This input is intermittent because the compressor is controlled by on-off

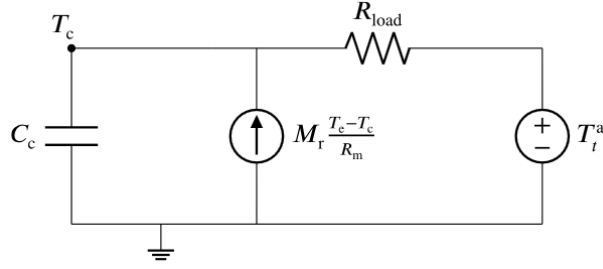


Figure 6.1.: RC-diagram of the model. The refrigerant mass state, M_r , is included in the heat source, illustrating the heat transfer to the cabinet. Source: [17].

logic. As a result, the cabinet model behaves similarly to a forced RC circuit, where the stored thermal energy changes due to heat gain from the ambient environment and heat removal by the refrigeration system.

The model simulates the internal cabinet temperature from three physical parameters: thermal resistance R , thermal capacitance C , and heat-removal coefficient G . Its inputs are the ambient temperature $T_a(t)$ and the compressor state $u(t)$. The output is the simulated internal temperature, which is compared with the measured internal temperature during parameter identification.

The electrical analogy leads to the following differential equations:

$$\frac{dq_c(t)}{dt} = -\frac{1}{RC}q_c(t) + I(t) + \frac{1}{R}T_a(t), \quad (6.1)$$

$$V_c(t) = \frac{1}{C}q_c(t), \quad (6.2)$$

where $q_c(t)$ is the analogous capacitor charge, $V_c(t)$ represents the cabinet temperature, R is the thermal resistance of the enclosure, C is the effective thermal capacitance of the cabinet contents, $I(t)$ is the analogous current representing heat removed by the evaporator, and $T_a(t)$ is the ambient room temperature.

For implementation, the model is written in state-space form:

$$\dot{x}(t) = Ax(t) + Bu(t), \quad (6.3)$$

$$y(t) = C_s x(t) + Du(t), \quad (6.4)$$

where $x(t)$ is the state, $u(t)$ is the input vector, and $y(t)$ is the simulated internal temperature. To avoid ambiguity with the thermal capacitance C , the output matrix is denoted as C_s . The matrices used in the model are:

$$A = -\frac{1}{CR}, \quad (6.5)$$

$$B = \begin{bmatrix} G & \frac{1}{R} \end{bmatrix}, \quad (6.6)$$

$$C_s = \frac{1}{C}, \quad (6.7)$$

$$D = \begin{bmatrix} 0 & 0 \end{bmatrix}. \quad (6.8)$$

The first input corresponds to the compressor state, scaled by the heat-removal coefficient G , and the second input corresponds to the ambient temperature contribution through the cabinet thermal resistance. The parameter vector estimated during training is therefore $P = [R, C, G]$. An example model simulation is shown in Figure 6.2.

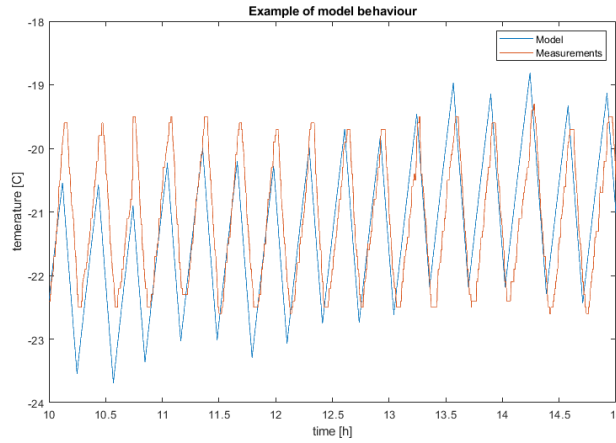


Figure 6.2.: Example of the model simulating the temperature inside a fridge. The data was collected over 5 h of normal freezer operation and includes ambient temperature T_a and compressor state as the model's inputs. We estimate R , C , and G .

This structure is broadly applicable to refrigeration appliances that consist of an insulated chamber, an ambient environment, and a heat exchanger. Its main advantage is that the fitted parameters remain physically interpretable: changes in R can be associated with insulation quality or door-opening disturbances, changes in C with the effective thermal mass inside the cabinet, and changes in G with the cooling effect produced during compressor operation.

6.3. Parameter Identification

The parameters are identified from a labeled training period containing internal temperature, ambient temperature, and compressor state. For each candidate parameter vector $P = [R, C, G]$, the state-space model is simulated, and its output is compared with the

measured internal temperature. The optimization objective is to minimize the difference between measured and simulated temperature, expressed by RMSE. In this implementation, the parameter fitting is performed with a genetic algorithm from MATLAB's optimization tools [21], following the approach used in [19].

6.4. Simulation with Hysteresis Controller

After parameter identification, the compressor state can be simulated using hysteresis logic. The model is given an ambient temperature profile, and the controller switches the simulated compressor on and off according to internal temperature thresholds. The resulting compressor state is then used to estimate on-time and energy consumption. This closed-loop simulation is important because replaying the measured compressor state is only useful for fitting, while prediction requires the model to generate compressor switching by itself.

6.5. Training and Testing Strategy

The training and testing strategy is designed to separate parameter identification from validation. Training data is used to fit the model parameters, while testing data is used to check whether the fitted model can predict operation outside the fitting period. The general workflow is:

1. Select a labeled training period.
2. Identify thermal model parameters.
3. Simulate future operation using ambient temperature and hysteresis control.
4. Compare the simulated compressor state with the measured compressor state on the test data.

Both sequential and random train/test splits may be considered. The convergence analysis conducted suggests a 30% training and 70% testing in several evaluations.

6.6. Convergence Analysis

A convergence analysis should determine how much training data is required before the gray-box model can be used for prediction. Figure 6.3 shows this analysis for Dataset 2, a freezer dataset with approximately one month of operation and 567 h of usable data. The MAPE decreases sharply when the training share is increased from 10% to 20%. Using 30% of the

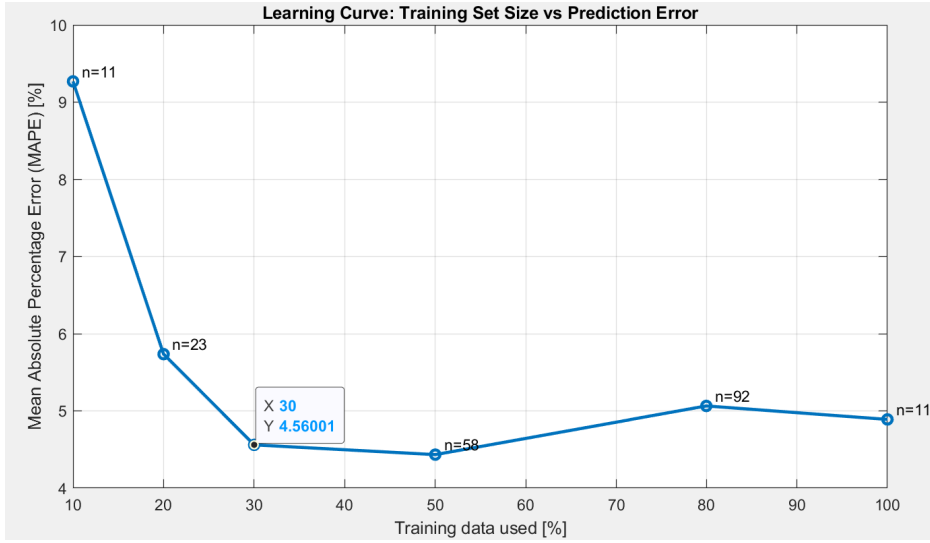


Figure 6.3.: Convergence of gray-box prediction error with increasing training-data share for Dataset 2.

dataset for training produces sufficiently low error, and adding more training data does not provide a meaningful further improvement.

6.7. Advantages and Limitations

The gray-box method requires more input information and labeled data than the peak-valley method, but it provides physically interpretable parameters and can simulate future operation under different ambient conditions. It may also support fault detection, predictive maintenance, and system identification. The main practical disadvantage is implementation complexity: parameter fitting requires dynamic simulation and optimization, which makes the method more computationally demanding than direct peak-valley detection.

7. Results and Comparative Evaluation

The experimental evaluation compares two ways of estimating compressor on-time, which is the main quantity required for simplified energy-consumption estimation. The peak-valley method is evaluated as a low-cost temperature-only estimator, while the gray-box method is evaluated as a more complex model-based estimator with additional diagnostic potential. The comparison focuses on accuracy, calibration or training requirements, practical deployment complexity, and possible use beyond direct energy estimation.

7.1. Evaluation Context

The results should be interpreted with the assumptions introduced in Chapter 3. The appliance power is approximated as a two-state function, where the dominant energy difference is between compressor-on and compressor-off operation. Therefore, the quality of energy estimation depends primarily on how accurately the compressor mask is reconstructed. The evaluation uses the MAPE of compressor on-time as the main metric. Lower MAPE indicates that the predicted compressor mask preserves the total duration of compressor operation in the analyzed segment.

The tested datasets differ in appliance type, duration, available measurements, and signal quality. This is important because both proposed methods use simplified assumptions. Detailed description of data used for experimental validation present in Chapter 3.3. The peak-valley method relies on the relationship between internal-temperature extrema and compressor switching. The gray-box method relies on a fitted physical model and therefore requires a labeled training period with compressor state, internal temperature, and ambient temperature. The results below should therefore be read not only as numerical accuracy values, but also as evidence of when each method is appropriate.

7.2. Peak-Valley Results

The peak-valley method estimates compressor operation from the shape of the internal-temperature signal. In the simplest case, detected temperature peaks and valleys are directly converted into compressor-on and compressor-off periods. When the extrema are shifted relative to the measured compressor transitions, a short calibration period can be used to estimate timing offsets.

Table 7.1.: Peak-valley compressor on-time estimation results.

Dataset	Appliance	Data used	Calibration	MAPE
1	Refrigerator	3.2 h	None	1.2%
2	Freezer	546 h usable	None	2.30%
3	Refrigerator	40 h usable	None	12.90%
3	Refrigerator	30 h test	First 30%	3.36%

Dataset 1 shows that high accuracy can be obtained directly from internal temperature for a short refrigerator measurement. Dataset 2 shows that the method can also remain accurate over long-term freezer operation without calibration. Dataset 3 shows the opposite case: without calibration, the timing mismatch between extrema and compressor transitions produced unacceptable error. After using the first 30% of the data for calibration, the error decreased substantially.

The results show that the peak-valley method can be highly accurate when the thermal response of the appliance produces extrema that are consistently related to compressor switching. Dataset 1 and Dataset 2 demonstrate that, in favorable conditions, only the internal chamber temperature is sufficient to estimate compressor on-time with low error. This is important from a deployment perspective because internal temperature is often already measured or can be added with a low-cost sensor.

Dataset 3 shows the main limitation of the method. Without calibration, the detected extrema did not align well enough with the real compressor transitions, producing 12.90% MAPE. This does not necessarily mean that the temperature signal contains no compressor-state information. Rather, it shows that the simple assumption of direct peak-to-valley mapping can fail when the thermal delay at compressor start-up and shutdown is not symmetric. After calibration, the error decreased to 3.36%, confirming that a short labeled period can make the method reliable when the timing offset is systematic.

The practical conclusion is that the peak-valley method is best treated as a low-cost estimator that should be calibrated when compressor-state data is available. After calibration, the method can potentially operate using only internal temperature. This makes it suitable for

embedded or edge applications where computational resources and sensor availability are limited.

7.3. Gray-Box Results

The gray-box method estimates compressor operation through a simplified physical model of the refrigeration appliance. In contrast to the peak-valley method, it requires a training period with internal temperature, ambient temperature, and measured compressor state. After parameter identification, the model can be used with hysteresis logic to simulate compressor switching and estimate compressor on-time.

Table 7.2.: Gray-box compressor on-time estimation results.

Dataset	Appliance	Usable data	Split	MAPE
2	Freezer	567 h	30% training, 400 h sequential test	5.2%
2	Freezer	567 h	Random 30–70 split	4.56%
3	Refrigerator	40 h	30 h test	4.95%

For Dataset 2, the sequential split tested whether the model could generalize from earlier operations to a later long testing period. The random split produced slightly lower error because training and testing data were more uniformly distributed across the available operating conditions. For Dataset 3, the gray-box method achieved accuracy comparable to the calibrated peak-valley method, but with a different set of requirements and capabilities.

The gray-box results are less dependent on direct peak and valley timing because compressor-state prediction is generated by a physical model and hysteresis controller. This makes the method conceptually more robust when the temperature extrema are shifted or when the thermal dynamics cannot be described by a simple signal-processing rule. The price for this robustness is the need for more measured signals and a more complex training procedure.

The gray-box method also has value beyond compressor on-time estimation. Since its parameters are physically interpretable, the same model can support system identification, anomaly detection, and predictive maintenance. For example, changes in thermal resistance may indicate insulation degradation, more frequent door openings, or other disturbances affecting heat transfer. This broader use case is one of the main reasons for including the gray-box method despite its higher implementation cost.

7.4. Comparative Discussion

Table 7.3 summarizes the most important practical differences between the two methods. The comparison shows that neither method is universally better. The peak-valley method is preferable when minimal hardware and low computational cost are the main requirements. The gray-box method is preferable when physical interpretability, simulation, or diagnostic capability is required.

Table 7.3.: Comparison of the two estimation methods.

Criterion	Peak-valley method	Gray-box method
Operation input	Internal temperature	Internal and ambient temp, compressor state
Labeled data	Calibration if needed	Training required
Computational cost	Low	Higher
Implementation	Simple signal processing	Dynamic model and optimization
Interpretability	Limited	Physical parameters
Future simulation	Limited	Supported
Diagnostics	Limited	Supported
Best use case	Low-cost monitoring	Model-based analysis

The comparison highlights an important trade-off. The peak-valley method is attractive because it can work with one low-cost temperature sensor and simple computation. This makes it suitable for retrofitting existing appliances, low-power monitoring devices, and embedded implementations. Its main risk is that signal extrema may not correspond directly to compressor transitions, especially when thermal delays differ between compressor start-up and shutdown. In such cases, calibration is necessary.

The gray-box method requires internal temperature, ambient temperature, and compressor state during training. Its implementation is more complex because it requires parameter identification, dynamic simulation, and compressor-control logic. The benefit is that the resulting model has physical meaning: the identified parameters can be interpreted as thermal resistance, thermal capacitance, and heat-removal coefficient. This makes the method useful not only for estimating compressor on-time but also for model-based diagnostics and prediction under changed operating conditions.

7.5. Practical Recommendations

For applications where the only goal is low-cost estimation of compressor on-time, the peak-valley method should be considered first. It has low hardware requirements, low computational cost, and can be implemented without a complex software environment. However, a short calibration period with measured compressor state is recommended whenever possible, because it can strongly reduce timing-related errors.

For applications where the appliance must be modeled, diagnosed, or simulated under different conditions, the gray-box method is more appropriate. It is more difficult to deploy, but it provides physically interpretable parameters and can support predictive maintenance and system identification. In practice, both methods can also be combined: the peak-valley method can provide a lightweight operational estimator, while the gray-box model can be used periodically or offline for deeper analysis.

8. Conclusions and Future Work

Compressor-state estimation from minimal sensor data can support energy-performance assessment when direct power measurements are unavailable or too costly to collect continuously. The thesis demonstrates that two different approaches are useful for this task: a simple peak-valley estimator intended for low-cost deployment, and a gray-box digital twin intended for physically interpretable modelling, simulation, and diagnostics. The main practical result is not that one method replaces the other, but that each method is appropriate under different sensor, computation, and deployment constraints.

8.1. Main Conclusions

This work presents two approaches to estimating the energy consumption of a refrigerator or freezer using minimal sensor data: the peak-valley method and the gray-box digital twin method. Both approaches estimate compressor operation first, and energy consumption is then obtained from compressor on-time and assumed power levels. The results show that internal temperature contains useful information about compressor operation, but the amount of additional data required depends on the chosen method and expected reliability.

The peak-valley method showed very good accuracy in favorable cases while requiring only the internal chamber temperature during normal operation. This is the main advantage of the method. It has low hardware requirements, low computational cost, and can be implemented in a simple embedded or edge environment, for example, in C on a low-power device. These properties make it suitable for low-cost monitoring systems and retrofitting cases where adding current or power sensors is undesirable.

The main limitation of the peak-valley method is that it depends on the relationship between temperature extrema and compressor switching. If the detected peaks and valleys are systematically shifted relative to the actual compressor transitions, the uncalibrated method can produce large errors. The results show that this problem can be mitigated with a calibration period containing compressor-state data. In Dataset 3, calibration reduced the MAPE from 12.90% to 3.36%, demonstrating that a short labeled period can make the method reliable when the timing error is systematic.

The gray-box method requires more information: internal temperature, ambient temperature, and compressor state during training. It also requires a more complex implementation, including dynamic simulation, parameter optimization, and hysteresis control. Training is computationally heavier than the peak-valley method and is more appropriate for a PC-class device using MATLAB, Python, or C++ rather than a simple embedded controller. For the largest dataset considered in this work, training can be on the order of an hour on a modern PC, depending on implementation and optimization settings.

The advantage of the gray-box method is that it is not only an estimator of compressor on-time. It is also a physical model of the appliance. This allows the model to predict operation under different conditions, such as different ambient temperatures, different thermal loads, or changes in thermal resistance caused, for example, by more frequent door openings or degraded insulation. In addition, the gray-box model can support anomaly detection, fault detection, predictive maintenance, and system identification, as demonstrated in related work on the same digital-twin modeling approach [19].

The overall conclusion is that the two methods serve different practical needs. The peak-valley method is the preferred solution when the goal is low-cost energy estimation with minimal sensing and minimal computation. The gray-box method is the preferred solution when the goal includes simulation, diagnostics, future-condition analysis, or physical interpretation of appliance behavior. Together, they provide a practical framework for energy-performance estimation at different levels of hardware availability and computational complexity.

8.2. Contributions

The thesis contributes a minimal-sensor framework for estimating compressor operation and energy performance in refrigeration appliances. The main contributions are summarized below.

- A peak-valley compressor-state estimation method based primarily on internal chamber temperature.
- A timing-calibration protocol that reduces systematic compressor-mask errors when a short labeled compressor-state period is available.
- A gray-box digital twin method for model-based compressor-state prediction using internal temperature, ambient temperature, and compressor-state training data.
- A comparative evaluation of both methods on measured refrigerator and freezer datasets.
- A practical discussion of the trade-off between low-cost deployment and physically interpretable model-based analysis.

These contributions are connected by the same practical goal: estimating energy consumption without requiring full power metering throughout the entire measurement period. This is important for applications where hardware cost, installation complexity, or data availability limit the use of direct electrical measurements.

8.3. Practical Benefits of the Proposed Approach

The practical benefit of the peak-valley method is that it reduces the sensing problem to a temperature-only estimation problem after calibration. Temperature sensors are cheap, small, and commonly available in refrigeration appliances. Because the algorithm is based on peak and valley detection plus simple timing operations, it can be deployed on low-power hardware and does not require a complex software stack.

The practical benefit of the gray-box method is different. It requires more input data and computation, but it produces a model that can be reused for analysis beyond the original measurement period. Once the model parameters are identified, the appliance can be simulated under different ambient-temperature profiles or under changed thermal assumptions. This makes the method useful for estimating future energy efficiency, evaluating the effect of changed operating conditions, and detecting behavior that is inconsistent with the expected physical model.

Using both methods gives flexibility. In a low-cost deployment, the peak-valley method can provide continuous compressor on-time estimation. In a more advanced diagnostic workflow, the gray-box model can be trained periodically or offline to identify physical changes in the system. This combination is useful because practical refrigeration monitoring often has conflicting requirements: low cost during operation, but richer interpretation when abnormal behavior is suspected.

8.4. Limitations and Future Work

The main limitation of both approaches is the simplified energy model. The thesis assumes that appliance power can be represented by two states: compressor on and compressor off. In real systems, auxiliary components such as fans, lighting, defrost heaters, controllers, and variable-speed compressors may contribute to power consumption. If these effects are significant, additional states or direct power measurements may be needed to improve the energy estimate.

Another limitation is the need for labeled data. The peak-valley method can operate on internal temperature alone after calibration, but the calibration itself requires compressor-state

data. The gray-box method requires compressor-state data during training. Future work should therefore investigate automatic or semi-supervised ways to reduce the dependence on labeled compressor-state measurements.

Future development should also include automatic estimation of P_{on} and P_{off} , door-opening detection, adaptive calibration for changing operating conditions, variable-speed compressor modeling, multi-compartment appliance modeling, and richer gray-box structures. Another useful direction is transferring identified parameters or calibration settings between appliances of similar type. This could reduce the amount of labeled data needed for new deployments and make the proposed methods more practical in large-scale monitoring systems.

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